

Ethics Perspectives and Cultural Differences in Engineering

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Outline

- Introduction
- Bad Examples of Ethics and Cultural Issues in Engineering
- Good Examples of Ethical and Culturally-Aware Engineering
- Conclusion

Introduction

- Ethics and cultural differences play an important role in engineering decisions.
- When communication, honesty or cultural awareness is missing, engineering failures can happen.
- I will show real examples of negative and positive cases where ethics and culture had a strong impact on engineering results.

Fukushima Nuclear Disaster Japan, 2011



- A major earthquake and tsunami struck Japan in 2011.
- The Fukushima nuclear plant lost power and the cooling systems failed.
- Three nuclear reactors melted down and released radiation.

Fukushima Nuclear Disaster Japan, 2011



- Engineers hesitated to question management due to strict hierarchy in Japanese culture.
- Safety warnings and risk reports were downplayed or ignored.
- Cultural silence and poor communication contributed to the disaster.

Boeing 737 MAX USA, 2018–2019



- Two Boeing 737 MAX aircraft crashed shortly after takeoff.
- Both accidents were caused by incorrect MCAS system activation.
- A total of 346 people died.

Boeing 737 MAX USA, 2018–2019



source: WGBH

- Boeing did not inform pilots about the MCAS system to avoid delays and extra training.
- Internal culture prioritized competition and speed over safety.
- Safety concerns were ignored, showing a failure in engineering ethics.

Volkswagen Dieselgate

Germany, 2015



source: The Atlantic

- Volkswagen installed software to cheat emissions tests in diesel cars.
- Cars passed laboratory tests but polluted far more in real driving.
- Millions of vehicles worldwide were affected.

Volkswagen Dieselgate Germany, 2015



source: The Now

- Engineers were pressured by a perfectionist corporate culture to achieve ideal results.
- The company deceived regulators and customers.
- This case shows how cultural pressure can lead to unethical engineering decisions.

Rana Plaza Factory Collapse Bangladesh, 2013



- A building with garment factories collapsed in Bangladesh.
- Workers had reported cracks in the walls the day before.
- More than 1,100 people died in the collapse.

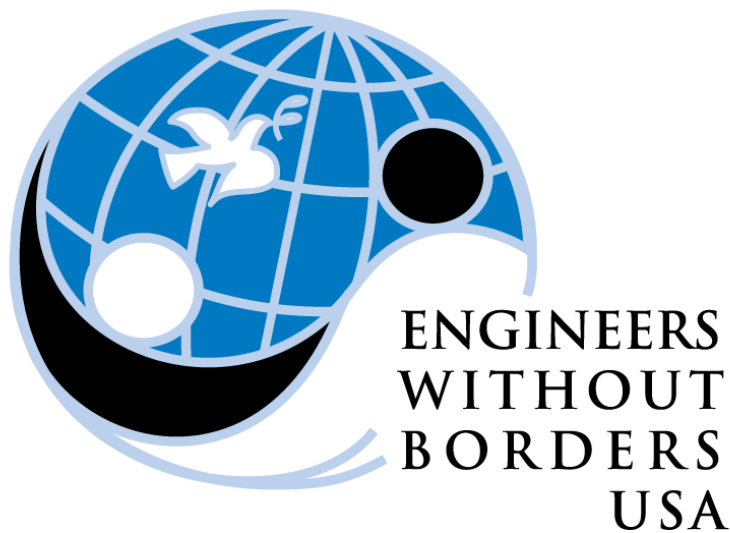
source: The New York Times

Rana Plaza Factory Collapse Bangladesh, 2013



- Workers were forced to enter the building due to local labor practices.
- Poor engineering standards and weak regulations contributed to the collapse.
- Cultural and economic pressure overruled safety and ethical responsibility.

Engineers Without Borders



source: Terrapin Works

- Engineers Without Borders develops sustainable engineering projects in developing countries.
- Projects include clean water, sanitation, housing and renewable energy.
- Engineers work directly with local communities.

Engineers Without Borders



source: The Borgen Project

- Solutions respect cultural habits, climate, and available materials.
- Ethical engineering is prioritized over profit.
- Projects empower communities and improve long-term safety and quality of life.

Toyota Kaizen Culture Japan



source: LinkedIn

- Toyota uses a “Kaizen” system, meaning continuous improvement.
- Every worker can stop production if they see a safety risk or defect.
- This approach has created one of the safest and most reliable engineering systems in the world.

Toyota Kaizen Culture Japan



- Japanese teamwork culture encourages responsibility and honesty.
- Safety and quality are more important than speed or cost.
- Cultural values directly improve engineering ethics and product reliability.

source: LinkedIn

Conclusions

- These examples show how ethics and cultural differences strongly influence engineering outcomes.
- When ethical principles or open communication are missing, the consequences can be serious.
- When culture supports safety, transparency and community, engineering becomes more responsible and effective.



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Thank you!